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A Vision Based Wearable Assistive Navigation System for Visually Impaired Individuals

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***Abstract*— Assistive technology plays a vital role in enhancing the independence and mobility of individuals with visual impairments, as navigating unfamiliar environments remains a significant daily challenge. This research presents the design of a light weight assistive wearable voice-controlled navigation device for the blind. Unlike other systems that depend on computer vision or external radio anchors, the present solution uses Raspberry Pi and LiDAR sensor to create real-time distance maps of immediate indoor environment. The device determines user's location based on comparing the real-time distance data with a previously recorded library of room configurations. To enable hands-free interaction, voice command is used by the user in combination with on-board VOSK speech recognition engine. After setting the target, a text-to-speech engine provides audio directions through the headphones. While navigation, the LiDAR and a set of front-facing ultrasonic sensors scan the environment for sudden obstacles. Prototype is lightweight, quick in responding to the voice command and accurate in identifying room signatures.**

Keywords*— Assistive Navigation, Wearable Robotics, Indoor Navigation, LiDAR Sensing, Obstacle detection, Voice Recognition, Human–Robot Interaction.

I. INTRODUCTION (*HEADING 1*)

As per current global statistics provided by the World Health Organization (WHO), more than two billion people suffer from some kind of visual impairment. Many among this number face movement constraints on a day-to-day basis. This calls for an immediate engineering requirement of developing devices that can help regain spatial awareness.[1]. The biggest challenge that comes the way of visually impaired people is traversing unfamiliar indoor environments such as campuses, hospitals, and transport terminals on their own. Since the absence of visual elements makes it extremely difficult for visually impaired persons to identify the spatial geometry of the room, identify hallways, and also avoid any obstacle, the problem of indoor navigation poses a significant hurdle compared to outdoor tracking which has been made easy by the use of GPS satellites.

To address these challenges, several assistive navigation technologies like walking sticks and navigation assistive devices have been developed; however, there are limitations when it comes to environmental awareness. Although walking sticks can be effective in detecting obstacles that are within reach, they are not able to provide any information regarding obstacles that are far from the individual or guidance on how to navigate to a certain place. While most smartphone applications used for navigation have visual interfaces that may not work well for blind individuals.

Recent developments in robotics, micro sensors, and embedded computing have paved the way for the construction of assistive devices that enhance the independence and mobility of visually impaired individuals. However, these devices provide insufficient information regarding the unstructured environment, and fail to assist users reach their destination. Furthermore, standard navigation devices like GPS-enabled mobile applications have been built for outdoor settings and do not work efficiently indoors due to signal attenuation.

The shortcomings of existing navigation systems highlights the necessity of an intelligent assistive mechanism that is able to sense its environment, detect obstacles, and enable interaction with the user in real-time. To improve the effectiveness of indoor navigation for visually impaired users, this paper presents an assistive indoor navigation mechanism that uses hand-free operation, along with LiDAR sensing and localization, and voice-based interaction for navigation. The proposed system works fully on board the Raspberry Pi, where the current location of the user is determined via comparison of real-life distance maps and localized maps of room layouts. The processing of verbal commands allows selecting the destination point and receiving navigational instructions.

II. RELATED WORK

Over the past decades, various forms of electronic travel aids have been developed to deal with indoor navigation of visually impaired persons. Shoval et al. proposed a sonar-based travel aid that demonstrated the effectiveness of combining active distance sensors with simple sound cues that would reduce collisions faced by visually impaired travellers considerably [2]. With speech recognition technology improving, natural language processing became one of the key components of assistive hardware. Studies by Li et al. showed that the use of natural communication through voice commands provides a much more intuitive way of interacting

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compared to button-based interfaces, thus leaving both hands free for a white cane or guide dog [3]. There have also been several investigations into indoor localization and positioning techniques for assistive navigation systems. Thrun et al. developed a set of probabilistic algorithms for localization using robots, showing that sensor information can provide accurate position estimation in indoor environments. This clearly shows the advantage of distance sensor in indoor localization independent of any outdoor infrastructure [4].The improvements in speech recognition technology have made user's interactions with the assistive system even better. Li et al. proved that voice-enabled systems can provide an effective and natural interaction between a user and robotic systems. Such kind of interaction does not require any physical control and is particularly convenient for visually-impaired people [5]. A wearable assistive navigation system involving real-time Semantic Visual SLAM and deep learning-based object detection and voice-guided navigation for visually impaired individuals; however, its performance is limited by computational complexity, dependence on indoor environments, and reduced robustness in dynamic or changing scenes [6].

Although there are numerous researches on using cameras or radio-anchor-based infrastructures have been developed for indoor positioning, such techniques suffer from high latency, fast battery drain, and expensive initial installation. The proposed system addresses these limitations through a novel approach that uses a purely independent method of lightweight spatial signatures mapping. Through the use of the spinning LiDAR and its comparison of the immediate environment distance maps with the light-weighted local structures database, the proposed system successfully enables room level tracking in a fully standalone manner on Raspberry Pi platform.

III. Design of Navigation Assistive device

The mechanical design of the wearable assistive navigation device was done via Autodesk Fusion 360 Computer Aided Design software. This design highlights the need for light weight construction, ergonomics, and efficient incorporation of the sensors as well as electronics to achieve practicality of use by the visually impaired users. The device itself is realized in a small wearable form which allows hands free navigation. It is protected in a casing which houses the processing and sensing parts.

The design of the physical housing of the assistive device was made in Autodesk Fusion 360, considering all efforts to make the device lightweight and convenient for everyday wearing. To be worn without the help of hands, we created a modular housing made of 3D printed Polylactic Acid (PLA) to cover up the internal components of the device. All the parts are carefully arranged around the Raspberry Pi 4B, the battery pack and soundcards. The LiDAR is installed at the very top of the casing and has an absolutely open cutout so that the spinning laser could scan the whole room in a 360-degree arc without interfering with the housing itself. The ultrasonic sensors are installed below the lidar, aiming forward in order to detect low obstacles. In such a way, the center of gravity of the device remains constant, thus not causing any inconvenience during the movement of the user. In order to ensure that the device can be effectively used on a day-to-day basis, the whole design of the wearable system was made with a view to ensuring its prolonged comfort and lack of physical stress. The final design includes the housing of the compact size measuring about 120 mm by 85 mm by 65 mm. Being mainly built from the lightweight polylactic acid (PLA), the weight of the whole assembly including the Raspberry Pi 4B, the Lidar A1 unit, the ultrasonic sensors, and the battery pack equals to around 380 grams. The housing is additionally fitted with an ergonomic mount of a chest harness that provides even distribution of the weight over the body area.

IV. System Architecture

The proposed wearable assistive navigation system is designed to support visually impaired individuals by providing reliable environmental perception and real-time navigation guidance. The proposed system integrates a lightweight wearable platform with voice-based interaction, enabling users to specify their destination through natural speech. Furthermore, the system continuously detects obstacles using onboard sensors to facilitate reliable and efficient navigation in unstructured indoor spaces. The proposed architecture is shown in Fig 1.

The proposed system employs Raspberry Pi as its main processing unit responsible for executing the navigation algorithms and coordinating communication between different hardware modules. The overall hardware architecture integrates sensing, user interaction, and output components to enable real-time navigation assistance for visually challenged users. LiDAR sensor performs the scanning of the surrounding environment and measure distances in multiple directions, thus enabling obstacle detection and environmental awareness. An ultrasonic sensor assists in detecting nearby obstacles at short distances, providing additional safety by identifying objects that may not be immediately detected. A USB microphone captures the voice commands from the user, allowing interaction with the system through speech recognition. Audio output (earphones) provides real-time navigation instructions and obstacle warnings using audio feedback generated through a text-to-speech engine. The system can be powered either through a laptop during development and testing or by a portable power bank for mobile operation, thus enabling flexibility and portability. The hardware components, including the LiDAR sensor, Raspberry Pi, and other modules are securely housed in a custom-designed 3D-printed wearable frame fabricated using lightweight PLA material. The frame offers support for hardware components while ensuring portability and ease of use during system operation.

Perception of the environment and obstacle detection take place using data received from the LiDAR sensor and ultra sonic sensor. The LiDAR constantly performs scanning of the environment from different directions and generates distance values. These distance values are analyzed to determine the presence of nearby obstacles as well as to compute distance profiles for localization purposes.

The voice command interpretation process uses the VOSK speech-to-text software. This interprets the voice commands

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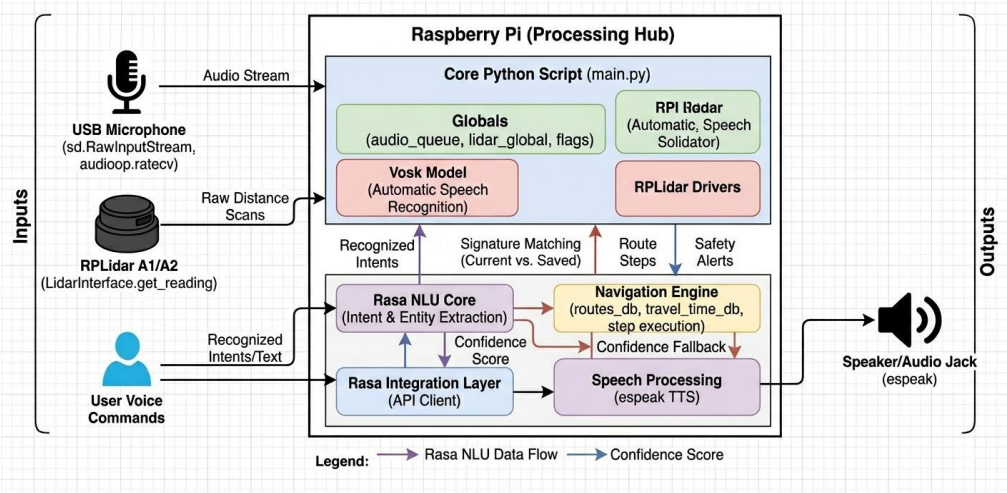


Fig. 1. System Architecture

that are picked up via the USB microphone into text. The interpreted voice commands are then used to find out where the user wants to go. The navigational prompts and alerts are communicated to the user via the audio feedback system which is done by the use of text-to-speech engine. This ensures that users who are visually challenged get a voice guide from the navigation system in a seamless way.

V. Methodology

The developed wearable assistive navigation device comprises sensing devices, voice input capability, and intelligent processing in order to provide safe navigation for the visually impaired. The system utilizes both hardware devices and software techniques in order to sense the environment, recognize any potential obstacles, recognize the location of the user, and generate auditory guidance

The general workflow of the proposed system involving the stages of system initialization, environmental sensing, sensor data processing, voice command recognition, location identification and navigation, and audio guidance is shown in Fig. 2.

A. System Initializtion

The Raspberry Pi takes on the role of the CPU and it initializes the operating system as well as all the hardware peripherals connected to the system. Hardware components like LiDAR sensors, ultrasonic sensors, USB microphones, and audio output modules are configured via communication ports like USB and GPIO. The navigation software is activated and the modules for sensor data collection, speech recognition, and navigation assistance are initialized.

B. Environmental Sensing

The system acquires continuous environmental information through its distance sensors. The LiDAR sensor sweeps around the environment and takes distance measurements from multiple angles, thereby making the system capable of detecting obstacles as well as analyzing the spatial arrangement of the environment. Further, the system has an ultrasonic sensor which is used to detect close range obstacles.

C. Sensor Data Procesing

Information about the environment is generated by pro cessing the raw data collected through sensors. Distance measurements are computed using LiDAR scan data in several directions. This distance measurement is further utilized to construct location signatures for identifying the distance pattern for different locations in the environment. Comparison between the stored location signatures and current readings provides the estimated position of the user.

D. Voice Command Recognition

The interaction between man and machine is achieved via voice commands. The voice command from the user is cap tured by the USB microphone. The VOSK speech recognition technology is used to process the speech command to produce text output. The text output produced above is interpreted to determine the destination or movement the user wants to do with the system. This makes it possible for the visually impaired to communicate conveniently with the system.

E. Location Identification and Navigation

After recognizing the user's voice command, the robot uses distance signatures based on LiDAR technology to determine its location and compare that to the already existing environmental data. Once the location is established, the navigation algorithm picks the best route to the destination point. While moving toward the destination, the system constantly checks the sensor data for any obstacles.

F. Audio Guidance and Feedback

Guidance and obstacle information will be conveyed to the user through audio feedback systems. The audio feedback system produces voice commands such as guidance or obstacle warning when there are obstacles in the surroundings. These audio guidelines are provided to the user through headphones that are plugged using an audio jack.

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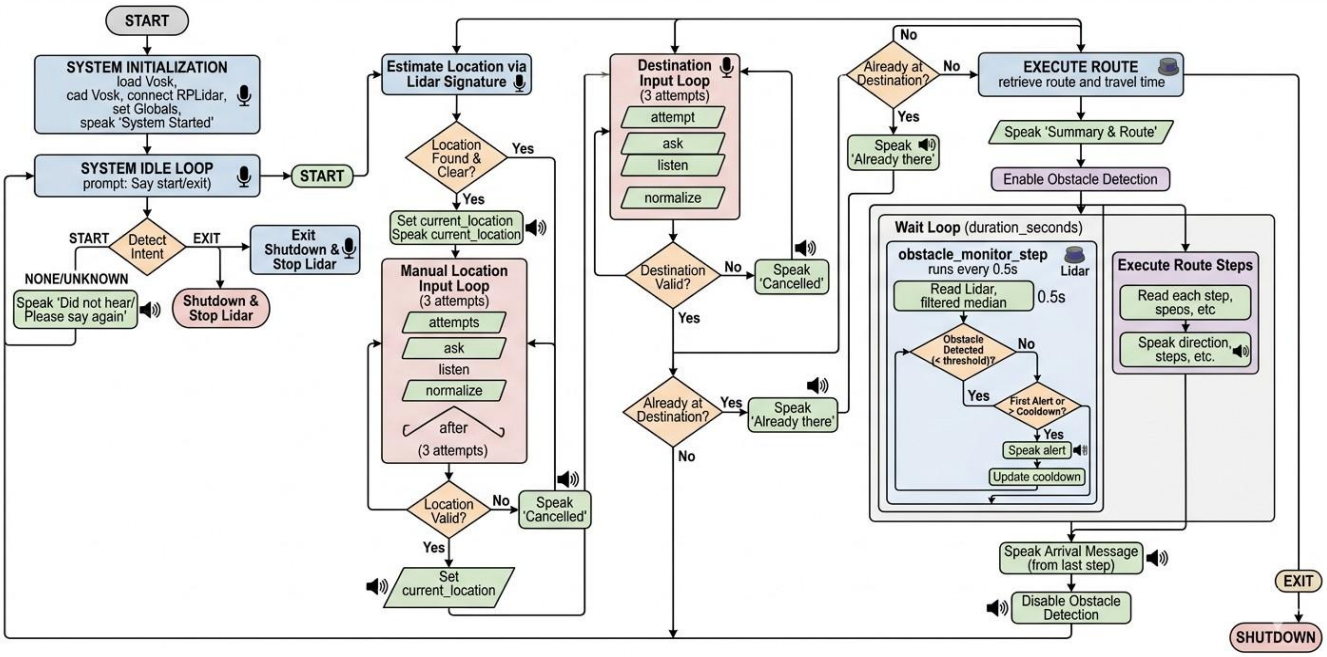


Fig. 2. Workflow of the wearable assistive navigation system

VI. Results and Discussions

The voice-based wearable assistive navigation system shown in Fig. 3, is tested in a controlled indoor environment that included the Automation Lab, Robotics Lab, ELDC Lab, and the male washroom.

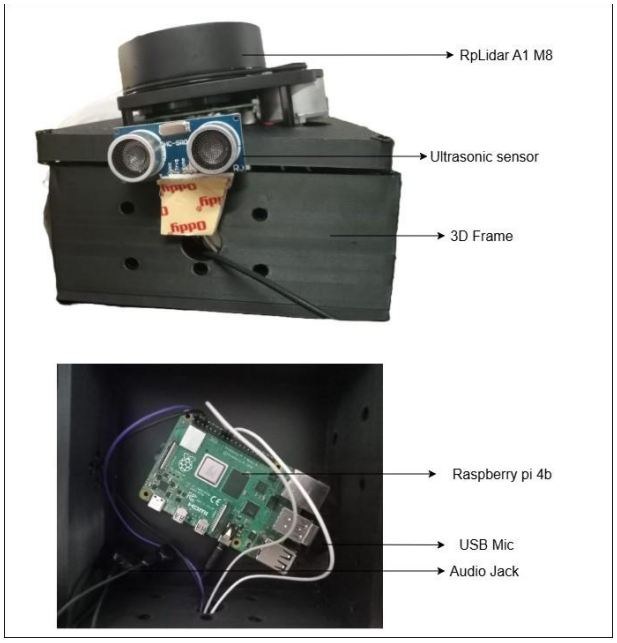


Fig. 3. Voice-Based Wearable Assistive Navigation System

The evaluation mainly focused on three key com ponents: voice command recognition, LiDAR-based localization, and real-time obstacle detection and navigation.

A. Voice Recognition Performance

The performance of the Vosk Small Model was evaluated under different noise conditions to assess its ability to recognize user voice commands accurately. The Speech Recognition Noise Performance Analysis is shown in Fig.4..

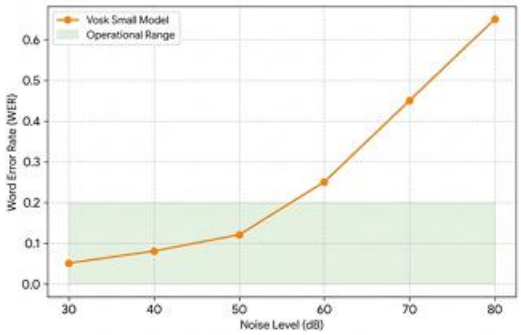


Fig. 4. Speech Recognition Noise Performance Analysis

As shown in Fig. 6, the Word Error Rate (WER) remains below 12% for noise levels ranging from 30 dB to 50 dB, which are representative of typical indoor laboratory environments. This demonstrates that the system provides reliable speech recognition under normal operating conditions. However, the recognition performance degrades significantly as the noise level exceeds 60 dB, with the WER increasing to approximately 65% at 80 dB. These results indicate that the proposed system performs effectively in quiet to moderately noisy environments but may require additional noise suppression or enhancement techniques for reliable operation in highly noisy environments, such as crowded public spaces or industrial settings.

B. Probabilistic Localization and Spatial Discrimination

The LiDAR-based localization system was evaluated for its ability to correctly identify and distinguish between

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rooms with similar layouts. The probabilistic heatmap of spatial localization confidence of the system is shown in Fig. 5.

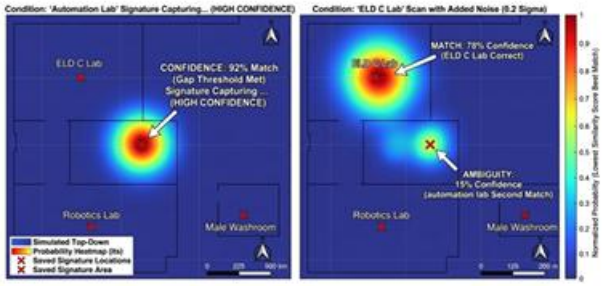


Fig. 5. Probabilistic Heatmap of Spatial Localization Confidence

The system achieved a high confidence level of 92% while identifying the Automation Lab under normal conditions, clearly separating it from other possible matches. To test robustness, Gaussian noise (= 0.2) was introduced to simulate environmental variations. In this case, the confidence dropped to 78%. Interestingly, the system was still able to recognize uncertainty by entering an "Ambi guity" state when the second-best match reached 15% confidence. This behavior highlights the effectiveness of the LOCATION SCORE GAP THRESHOLD in handling uncertain scenarios.

C. *Obstacle Detection and Navigation*

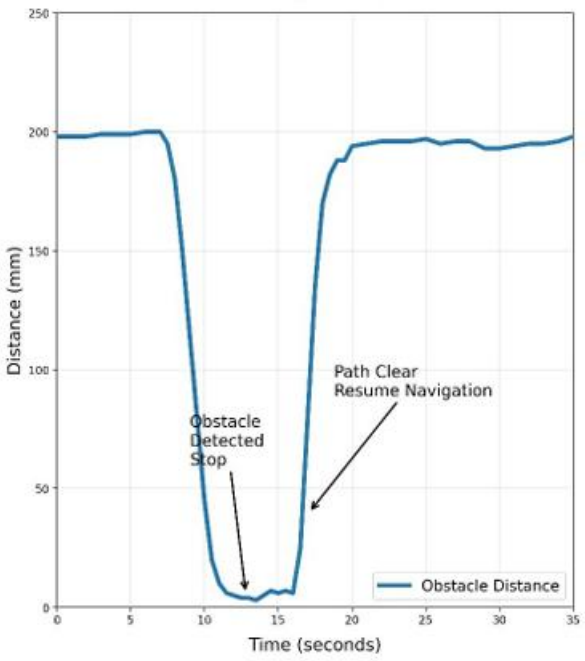


Fig. 6. Ultrasonic Sensor Readings During Navigation Showing Obstacle Detection And Clearance.

The obstacle detection performance of the ultrasonic sensor is shown in Fig 6 .The sensor continuously monitors distances to nearby objects. During one test, an obstacle was introduced at the 10-second mark. The system quickly detected a sudden drop in distance to approximately 3 mm,triggered an "Obstacle ahead" voice alert, and imme diately stopped for safety. Once the obstacle was removed at the 17-second mark, the system resumed navigation smoothly, maintaining accurate route guidance.

The overall performance of the proposed wearable assistive navigation system was evaluated based on localization accuracy, obstacle detection range, and voice command recognition performance. The summarized experimental results are shown in Table I.

TABLE I. SYSTEM PERFORMANCE RESULTS

Module	***Observation***	***Result***
Localization	Successful LiDAR localization	Automatic detection
Voice Interface	Commands recognized	Successful
Navigation	Path execution	Accurate
Obstacle Detection	Detected at ~3 cm	Safe stop achieved
System Completion	Destination reached	Successful

VII. LIMITATIONS AND FUTURE WORK

The identification of locations using LiDAR distance signatures can be influenced by any changes in the environment, including presence of other objects or changed environment itself. Also, processing capabilities of the Rasp berry Pi could affect performance in case if sensing and voice recognition modules work together. Currently, testing was performed only within indoor settings, so additional testing in different conditions should be done. Further research would involve enhancing the accuracy of localization and navigation skills using more sophisticated techniques like Visual SLAM and ROS 2 Navigation stack. Improvements can further involve the application of other types of sensors, improved obstacle avoidance algorithms, and perception using deep learning. Real-world testing with blind individuals will also be carried out extensively to increase system usability and reliability.

Although the preliminary bench testing had proven to be successful in confirming the low latency of the audio feedback loop and accuracy of our room signature matching, the process of validating any kind of assistive technology would require feedback from the visually challenged people. This is addressed by establishing a user-safety trial frame work in our next phase of research. The trial will be done using a qualitative evaluation matrix that centers on three essential measures: audio clarity of instructions, real-time safety of ob stacle avoidance, and physical comfort of wearing. Firmware improvements to the system, including timing thresholds of the navigation alerts from text-to-speech software, will be fine-tuned according to user walking speed and orientation.

VIII. CONCLUSION

This paper described a wearable assistive navigation aid that would allow visually challenged people to navigate indoors independently. The proposed device incorporates environmental scanning using a LiDAR sensor, obstacle detection using ultrasonic sensors, and human-machine interaction through voice commands in an embedded system built on Raspberry Pi. Experiments conducted show that using both technologies for sensing and speech recognition leads to successful detection of obstacles and navigation aid. The system is able to recognize user's commands, detect obstacles in real time, and give voice navigation instructions between various points. The results indicate that such assistive navigation systems have strong potential to enhance

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environmental awareness and improve mobility for visually challenged individuals in indoor environments.

REFERENCES

[1] WHO, "Blindness and visual impairment," 2026. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment

[2] Shoval, J. Borenstein, and Y. Koren, "Mobile Robot Obstacle Avoidance in a Computerized Travel Aid for the Blind," in *Proc. IEEE Int. Conf. Robotics and Automation*, 1994.

[3] A. Farghal, I. M. Kamal, and M. Salah, "Camera-Based Navigation System for Blind and Visually Impaired People," *Sohag Engineering Journal*, vol. 2, no. 1, pp. 1–8, 2022.

[4] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. Cambridge, MA, USA: MIT Press, 2005.

[5] S. A. Li, Y. Chen, and J. Wang, "Voice Interaction Recognition Design in Real-Life Scenario Mobile Robot Applications," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, 2023.

[6] Z. Chen, X. Liu, M. Kojima, Q. Huang, and T. Arai, "A Wearable Navigation Device for Visually Impaired People Based on Real-Time Semantic Visual SLAM," *Sensors*, vol. 21, no. 4, p. 1536, 2021.

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